

## Max-Margin classifiers or Support vector machines.

Consider a Binary classification problem.

$$x \in \mathbb{R}^d \quad \& \quad y \in \{-1, 1\}$$

Discriminant functions:

Recall that, a binary classifier can be represented as,

$$h(x) = \begin{cases} 1 & g(x) > 0 \\ -1 & \text{otherwise.} \end{cases}$$

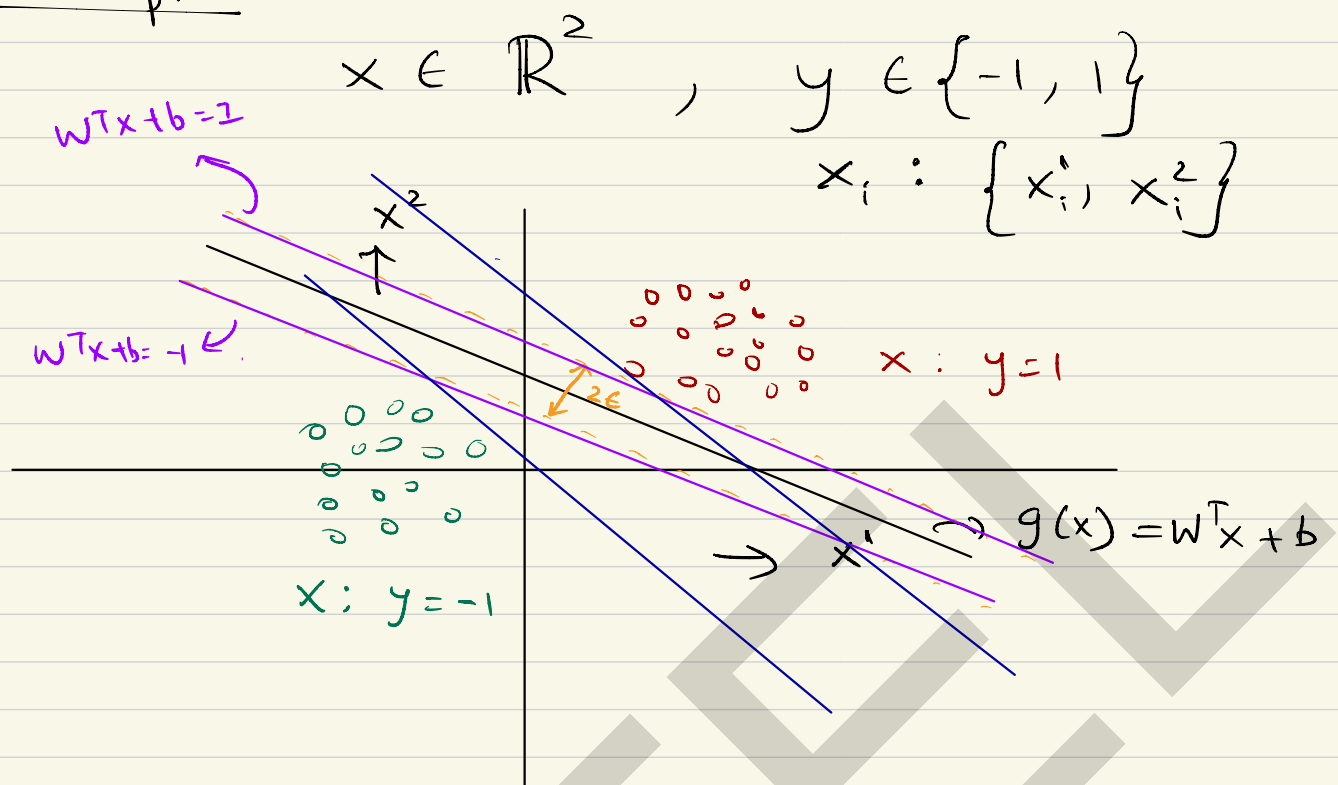
where  $g(x)$  = function of  $x$ .

$$g(x) : \mathbb{R}^d \rightarrow \mathbb{R}.$$

$g(x)$  : discriminant function.

For some specific cases of class-cond. densities,  $g(x)$  will be linear.

Example:



Linear Separability: A Dataset  $D$  is

linearly separable iff  $\exists w \& b$

s.t  $w^T x_i + b > 0 \quad \forall x_i : y_i = 1$

$w^T x_i + b < 0 \quad \forall x_i : y_i = -1$

Linearly separable data will have infinitely many discriminant functions.

question: Is there an "optimal" discriminant fn. among all these?

## Optimality via maximizing the margin.

Consider lin-separable data.

Since it is lin-separable,  $\exists \epsilon > 0$  s.t

$$W^T x_i + b \geq \epsilon \quad \forall x_i : y_i = 1$$

$$W^T x_i + b \leq -\epsilon \quad \forall x_i : y_i = -1$$

Now,  $W$  &  $b$  can be rescaled, such that

$$\hat{W}^T x_i + \hat{b} \geq 1 \quad \forall x_i : y_i = 1$$

$$\hat{W}^T x_i + \hat{b} \leq -1 \quad \forall x_i : y_i = -1$$

The above can be combined as

$$y_i (W^T x_i + b) \geq 1 \quad \forall i.$$

For lin. sep. case,  $\exists W$  &  $b$  s.t

$$y_i (W^T x_i + b) \geq 1 \quad \forall i$$

also, there exists no datapoint

blw the hyperplanes,  $W^T x + b = 1$  &  $W^T x + b = -1$   
(margins)

& they are parallel to  $W^T x + b = 0$

Consider the distance (perp.) b/w the margins:  $\frac{2}{\|W\|}$

one criteria for optimality of  $g(x)$  is that  $\frac{2}{\|W\|}$  has to be maximized.

Mathematically,

$$\begin{aligned} \min_w \quad & \frac{\|W\|^2}{2} \\ \text{s.t.} \quad & (W^T x_i + b) y_i \geq 1 \quad \forall i \\ & [1 - (W^T x_i + b) y_i] \leq 0 \end{aligned}$$

The above problem is a constrained optimization problem, with convex cost & linear constraints.

Solve the above via KKT optimizations.

Define Lagrangian for the above problem:

$$L(w, \mu, b) = \frac{w^T w}{2} + \sum_{i=1}^n \left[ 1 - y_i (w^T x_i + b) \right] \mu_i$$

KKT for the above problem :

$$a) \nabla_w L(w^*, \mu^*) = 0$$

$$b) \mu_i^* \geq 0$$

$$c) \mu_i^* \left( 1 - y_i (w^T x_i + b) \right) = 0$$

$$a) \nabla_w L = 0 \Rightarrow \boxed{w^* = \sum_{i=1}^n \mu_i^* y_i x_i}$$

$$\nabla_b L = 0 \Rightarrow \sum_{i=1}^n \mu_i^* y_i = 0$$

$$b) \mu_i \geq 0 \quad \forall \quad i$$

$$\text{Define } S = \{x_i : \mu_i^* > 0\}$$

$$\Rightarrow w^* = \sum_S \mu_i^* y_i x_i$$

To get  $w^*$ , we need the optimal lag. multipliers  $\mu_i^*$ .

To find  $\mu_i^*$ , we solve the dual problem

Duality of optimization:

$$L(w, \mu) = \frac{w^T w}{2} + \sum_{i=1}^n \left[ 1 - y_i (w^T x_i + b) \right] \mu_i$$

Define a Dual function  $q: \mathbb{R}^n \rightarrow \mathbb{R}$

$$q(\mu) = \inf_w L(w, \mu)$$

↪ greatest lower bound.

Construct the Dual optimization problem as follows:

$$\begin{aligned} \max_{\mu} \quad & q(\mu) \\ \text{s.t.} \quad & \mu_i \geq 0 \quad \forall i \end{aligned}$$

primal-dual relationship:

$w^*, \mu^*$  are optimizer for primal & dual respectively iff

i)  $w^*$  &  $\mu^*$  are feasible for primal & dual respectively.

$$\text{ii) } L(w^*, \mu^*) = \min_w L(w, \mu^*)$$